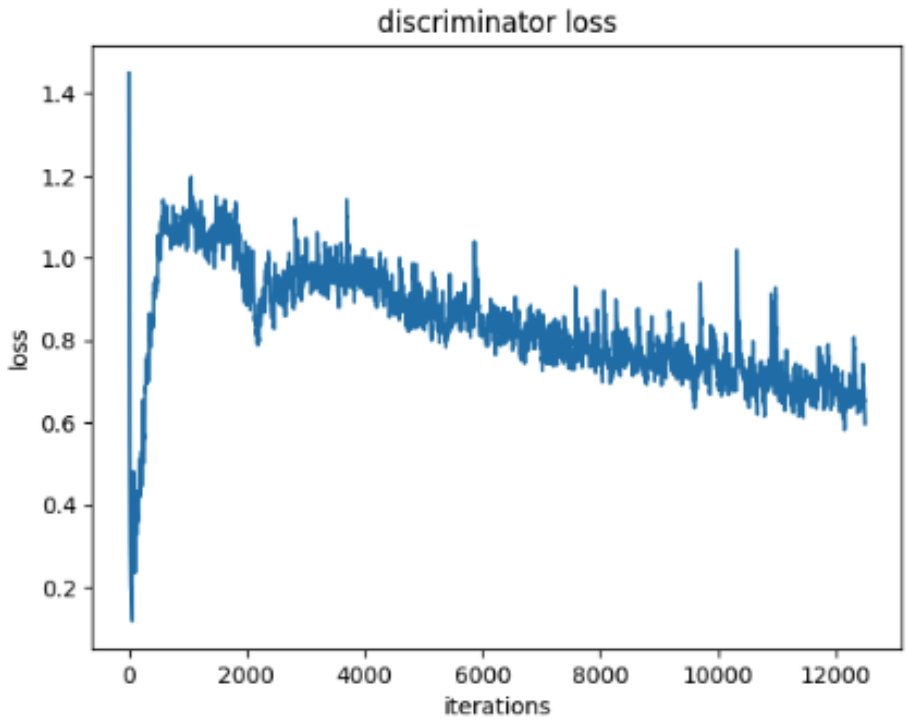
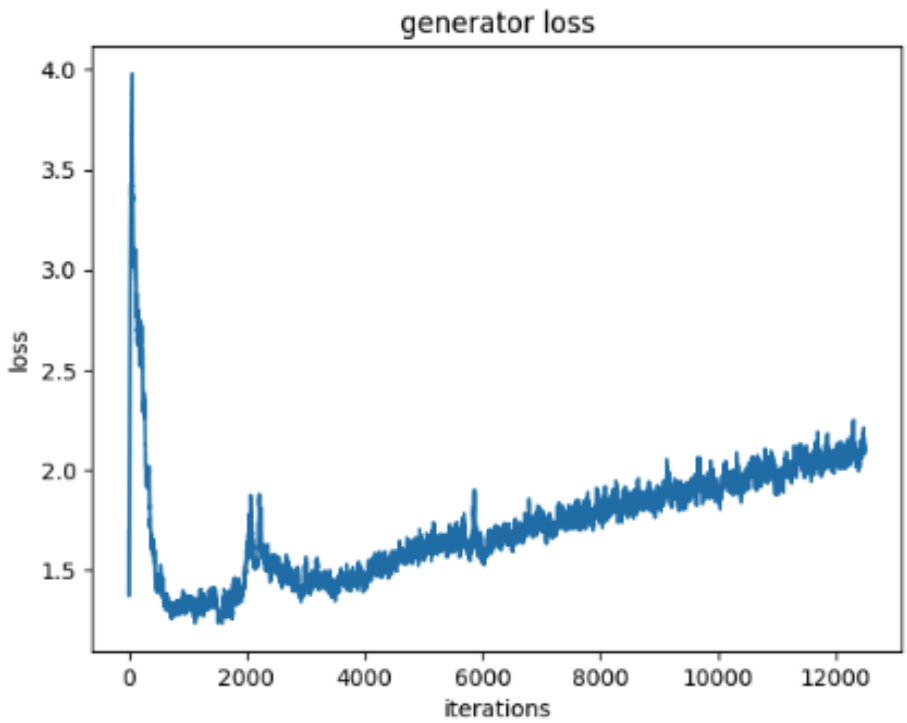
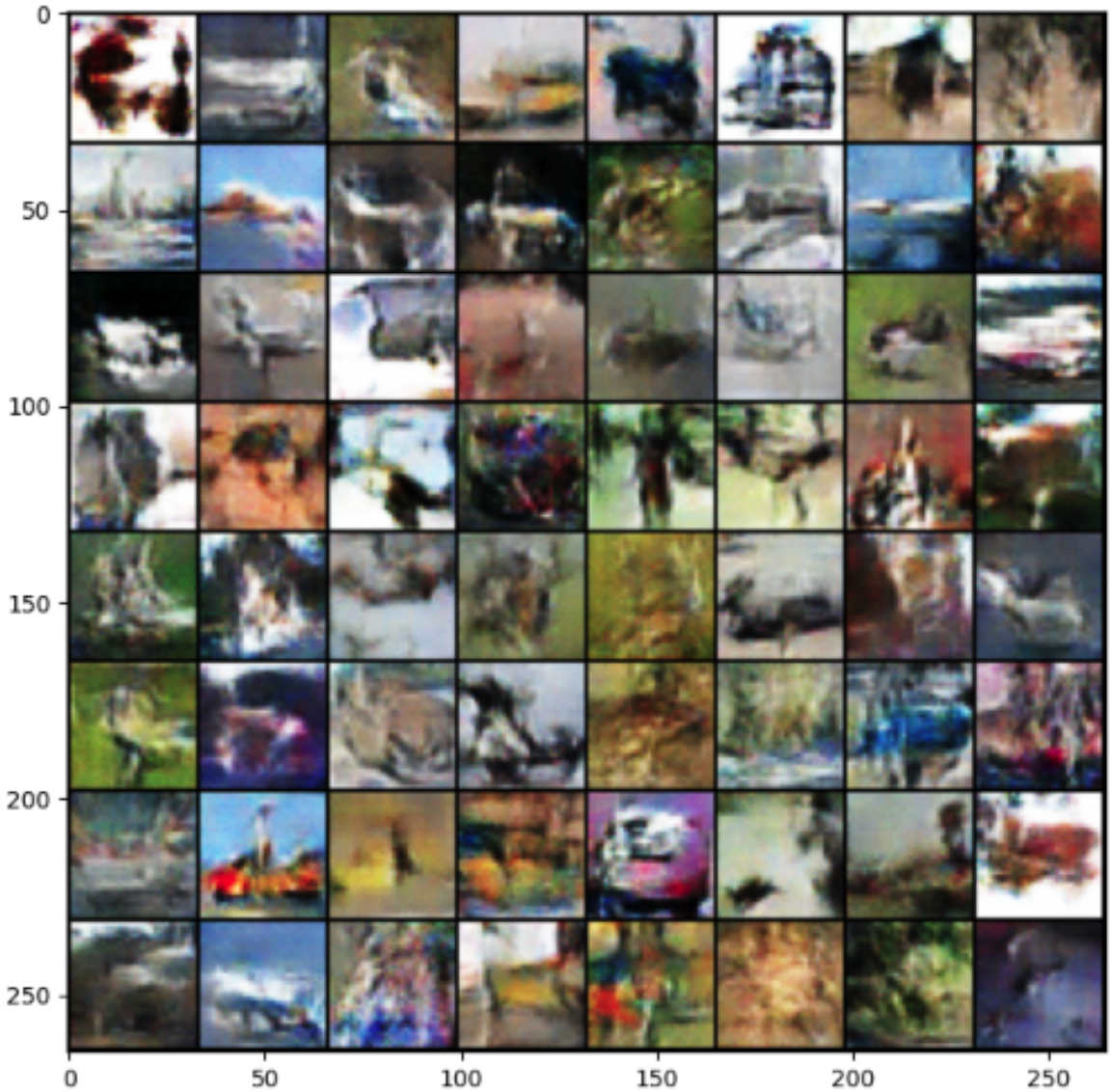
# Problem 2-1: Training Curves and Generated Images (20 pts)

## Loss Curves



### Generated Images at End of Training




## Problem 2-2: The Forger vs. the Police, Revisited (2 pts)

**Question.** In the forger-versus-police story, when we consider one of the two parties, the other is treated as a black box: they know their opponent's *result* but not *how* it works. What is wrong with this as an analogy for GAN training?

**Answer.**

The problem with this analogy is that it treats the opponent as a black box. This means each side only sees the other's output and does not know what is happening inside.

But this is not true in a real GAN. When training the generator, its output goes through the discriminator, and gradients are passed back through it.

So, the generator does not just see the final output. It also receives information from inside the discriminator through gradients.

Because of this, the two models are not simply observing each other's outputs. The generator actually uses internal information from the discriminator, so the analogy is not accurate.

## Problem 2-3: The Batch Normalisation Dilemma (5 pts)

**Question.** By removing the first BatchNorm layer in the discriminator, the only remaining failure case is when two distributions produce post-`LeakyReLU` activations such that one is an isotropic scaling and translation of the other — an extremely unlikely coincidence. Propose a *different* feeding strategy that solves the original BatchNorm problem *without* removing any BatchNorm layers or changing their mode of operation.

**Answer.**

A possible solution is to always feed the discriminator with mixed batches consisting of both real and generated samples. Specifically, at each training step, each batch is constructed such that half of the samples are real and the other half are generated.

In this way, the mean and variance used in Batch Normalization are always computed from the same mixed distribution (real + fake). As a result, the normalization process remains consistent between the discriminator training step and the generator training step.

This eliminates the inconsistency where generated samples are normalized differently across the two phases, and helps prevent the generator from learning an incorrect distribution.

## Problem 2-4: Activation Maximisation (10 pts)

### Activation Maximisation

## Image Reconstruction

**Average reconstruction loss** = 0.0116

Figure 1: Original

Figure 2: Reconstructed